Albin Zeqiri

PhD Candidate in Human–Computer Interaction, Ulm University

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Profile

I am a **final-year PhD candidate in Human–Computer Interaction** at Ulm University. Using my background in machine learning and experimental research methods, **I study how interactive systems can help users reach their intended goals while meeting ecological objectives set by companies and policymakers**. Leveraging a use-case-based research approach, I have studied this across contexts such as mobility, appliances, and urban robotics. My current work focuses on building design, where I use XR and generative AI to support the planning and use of spaces. In collaborations across industry and academia (**Mercedes-Benz Tech Innovation, UCL Interaction Centre, Karlsruhe Institute of Technology**), I was responsible for planning, implementing, and evaluating customized machine learning pipelines. My research and collaborative projects have been published at top-tier venues such as **CHI** and **IMWUT**.

Skills

Experimental Design: Online, laboratory, and field-based experiments (exploratory and hypothesis-driven); large-scale literature surveys (e.g., PRISMA); data mining using multi-API data collection and NLP-based analysis
Analysis: Statistical analysis; machine learning and deep learning (supervised and self-supervised); qualitative analysis using thematic analysis and grounded theory
Generative & Agentic AI: LLM APIs (OpenAI, Anthropic, Llama); LLM agents with tool use; diffusion models (Stable Diffusion, ComfyUI); agentic coding (Claude Code, Cursor, GitHub Copilot)
Programming Languages: Python, C#, JavaScript, R, SQL, C/C++, Java, Bash, HTML, CSS
Frameworks and Tools: PyTorch, TensorFlow, CUDA, OpenCV, Unity, React, Node.js, Flask, Git, PostgreSQL
Design: UI/UX prototyping and wireframing; image and video editing (Adobe Creative Cloud)
Languages: Albanian (native), German (native), English (fluent), French (intermediate)

Work Experience

Ulm University – Chair of Human–Computer Interaction 09/2022 – present
Research Associate

- Studied how interactive systems can meet ecological goals without hurting user experience through experiments, field deployments of artifacts, and interviews, resulting in **three first-author full papers at CHI and IMWUT**.
- Independently initiated and maintained a collaboration with the Botanical Garden Ulm, leading the development of a robot-based remote-gardening system and a mobile visitor app to evaluate design concepts with real users (**CHI '25, CHI '26**).
- Built reproducible deep learning pipelines, ablation studies, and multi-API data mining workflows for **UCL Interaction Centre**, **KIT**, and **Mercedes-Benz Tech Innovation**, leading to three joint publications.
- Co-authored a successful **€1,250,000 DFG Reinhart Koselleck proposal** on everyday XR for better building utilization, which frames my current building-design work.
- Supervised 40+ students from problem formulation to publication; **two student projects were published at CHI** and two students continued to a PhD.

Ulm University – Chairs of Human–Computer Interaction and Visual Computing 12/2019 – 04/2022
Research Assistant and Tutor

- Built VR prototypes in Unity and C# and supported the user studies run with them, giving the group working testbeds for new interaction techniques.
- Benchmarked monocular depth estimation models for accuracy and scalability to inform model selection.
- Taught C#, Unity, and UI/UX prototyping in tutorials.

Education

PhD in Human–Computer Interaction 09/2022 – 12/2027 (expected)
Ulm University
Dissertation: Between User Experience and Planetary Limits: Designing Interactive Systems with Ecological Objectives
Advisor: Prof. Dr. Enrico Rukzio

M.Sc. Computer Science 03/2020 – 08/2022
Ulm University Grade: 1.3 (A-equivalent)
Thesis (1.0, A-equivalent): A Dataset and Temporal Modeling Approach for Automated Thermal Comfort State Recognition, **published in IMWUT '24** (doi:10.1145/3678503)

B.Sc. Computer Science 10/2016 – 12/2019
Ulm University Grade: 2.0 (B-equivalent)
Thesis (1.0, A-equivalent): Depth Levels: Measuring Achievable Levels of Voluntary Vergence Eye Movements for Eye-based Human–Computer Interaction

Selected Publications

CHI '26 **Investigating the Effects of Eco-Friendly Service Options on Rebound Behavior in Ride-Hailing**
A. Zeqiri, M. Rietzler, E. Rukzio · doi:10.1145/3772318.3790711
Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26)

CHI '25 **PlantPal: Leveraging Precision Agriculture Robots to Facilitate Remote Engagement in Urban Gardening**
A. Zeqiri, J. Britten, C. Schramm, P. Jansen, M. Rietzler, E. Rukzio · doi:10.1145/3706598.3713180
Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)

CHI '25 **Bumpy Ride? Understanding the Effects of External Forces on Spatial Interactions in Moving Vehicles**
M. Sasalovici, **A. Zeqiri**, R. C. Schramm, O. J. Ariza Nuñez, P. Jansen, J. P. Freiwald, M. Colley, C. Winkler, E. Rukzio · doi:10.1145/3706598.3714077
Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)

IMWUT '24 **'Eco Is Just Marketing': Unraveling Everyday Barriers to the Adoption of Energy-Saving Features in Major Home Appliances**
A. Zeqiri, P. Jansen, J. O. Rixen, M. Rietzler, E. Rukzio · doi:10.1145/3643558
Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT), Vol. 8, No. 1

IMWUT '24 **AutoTherm: A Dataset and Benchmark for Thermal Comfort Estimation Indoors and in Vehicles**
M. Colley*, S. Hartwig*, **A. Zeqiri**, T. Ropinski, E. Rukzio · doi:10.1145/3678503
Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT), Vol. 8, No. 3

* equal contribution. Full list on Google Scholar.

Fellowships & Grants

DFG Reinhart Koselleck Project *VRooms* (€1,250,000), co-author 2025
Co-authored the successful proposal with Prof. Dr. Enrico Rukzio: *Fighting Climate Change by Increasing the Utilization of Buildings through Everyday Extended Reality* (project page)

Landesgraduiertenförderung (LGFG) Doctoral Fellowship 2023
Competitive three-year fellowship funding my PhD research

DAAD Travel Grant 2026
German Academic Exchange Service grant to present at CHI '26

Graduate Travel Grant (€2,800) 2024
Graduate & Professional Training Center Ulm grant to present two papers at UbiComp '24